Meridian AP FX Exposure Bot

BC3415 Week 11 · RPA project · TagUI v6 · Jonathan Low · ECB rate date 2026-10-23

1. Summary

An RPA bot that does a finance team's morning foreign-exchange routine: it reads the European Central Bank's reference rates from the ECB website, keeps a daily rate history, converts every open foreign-currency supplier invoice into SGD, measures the FX risk, raises alerts and saves screenshots as audit evidence. It is built on idea 2 from the brief ("scrape daily FX rates and append to a CSV") and carried through to the accounts-payable task that the rates are needed for.

Invoices processed	Currencies read live	Open AP in SGD	FX risk at 1-month high
10	**6 + SGD**	**229,056.96**	**837.54**

Run time	Check vs ECB XML
81–86 s	PASS

2. Business problem

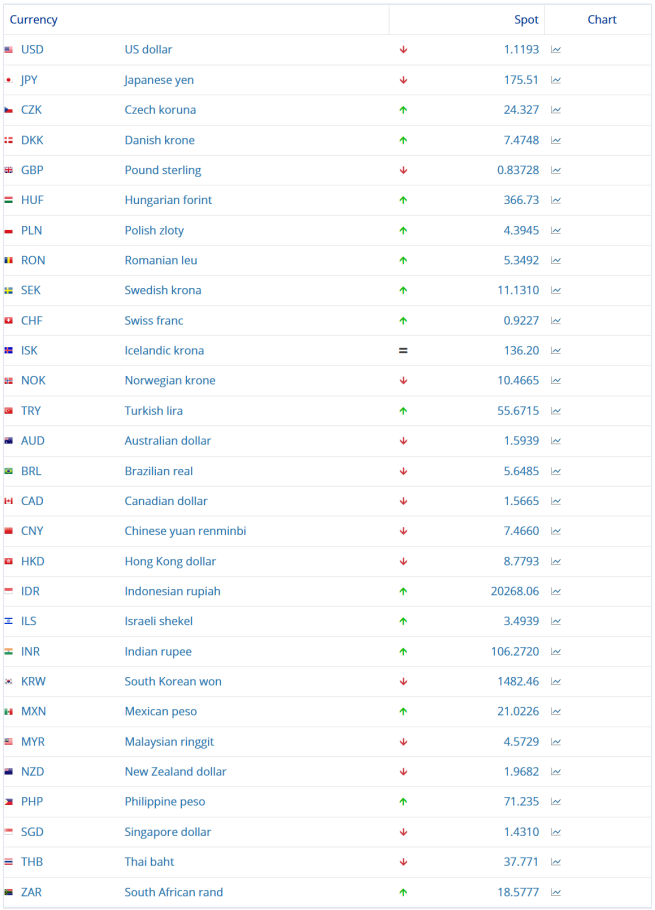
Meridian (the fictional company used in the Week 10 n8n lab) buys from suppliers in Europe, the US, Japan, Malaysia, China, the UK and Australia, and reports in SGD. Before each payment run an AP analyst:

1. opens the ECB reference-rate page and copies the rate for each currency into a spreadsheet;
2. works out SGD cross rates (the ECB quotes everything against the euro);
3. converts each open invoice, sorts them by due date and spots overdue ones;
4. checks whether any currency has moved sharply, which changes how much SGD to set aside;
5. saves a screenshot of the source page so auditors can see which rates were used.

That takes roughly 20–30 minutes a day, and copy-typing rates is a known source of errors (a JPY rate of 0.008153 is easy to mistype). It is rule-based, repetitive, done in a browser and a spreadsheet, and has no judgement in it until the exceptions: a good fit for RPA.

	Manual (as-is)	**Bot (to-be)**
Time per day	about 20-30 min of analyst time (estimate)	about 82 s unattended (measured)
Rates	typed by hand	read from the page's own elements
Audit evidence	screenshot if remembered	rate table, page and every chart captured each run
History	spreadsheet tab, if kept	CSV appended once per ECB date (re-runs safe)
Exceptions	noticed or not	unknown currency, overdue invoice and FX moves written to an alerts file
Who decides	analyst	analyst: the bot prepares numbers, it approves nothing

3. How the bot works



Currency			Spot	Chart
USD	US dollar	↓	1.1193	
JPY	Japanese yen	↓	175.51	
CZK	Czech koruna	↑	24.327	
DKK	Danish krone	↑	7.4748	
GBP	Pound sterling	↓	0.83728	
HUF	Hungarian forint	↑	366.73	
PLN	Polish zloty	↑	4.3945	
RON	Romanian leu	↑	5.3492	
SEK	Swedish krona	↑	11.1310	
CHF	Swiss franc	↑	0.9227	
ISK	Icelandic krona	=	136.20	
NOK	Norwegian krone	↓	10.4665	
TRY	Turkish lira	↑	55.6715	
AUD	Australian dollar	↓	1.5939	
BRL	Brazilian real	↓	5.6485	
CAD	Canadian dollar	↓	1.5665	
CNY	Chinese yuan renminbi	↓	7.4660	
HKD	Hong Kong dollar	↓	8.7793	
IDR	Indonesian rupiah	↑	20268.06	
ILS	Israeli shekel	↑	3.4939	
INR	Indian rupee	↑	106.2720	
KRW	South Korean won	↓	1482.46	
MXN	Mexican peso	↑	21.0226	
MYR	Malaysian ringgit	↓	4.5729	
NZD	New Zealand dollar	↓	1.9682	
PHP	Philippine peso	↑	71.235	
SGD	Singapore dollar	↓	1.4310	
THB	Thai baht	↓	37.771	
ZAR	South African rand	↑	18.5777	

Step 2-3: the ECB rate table, captured by the bot (snap of //table[@class='forextable']).

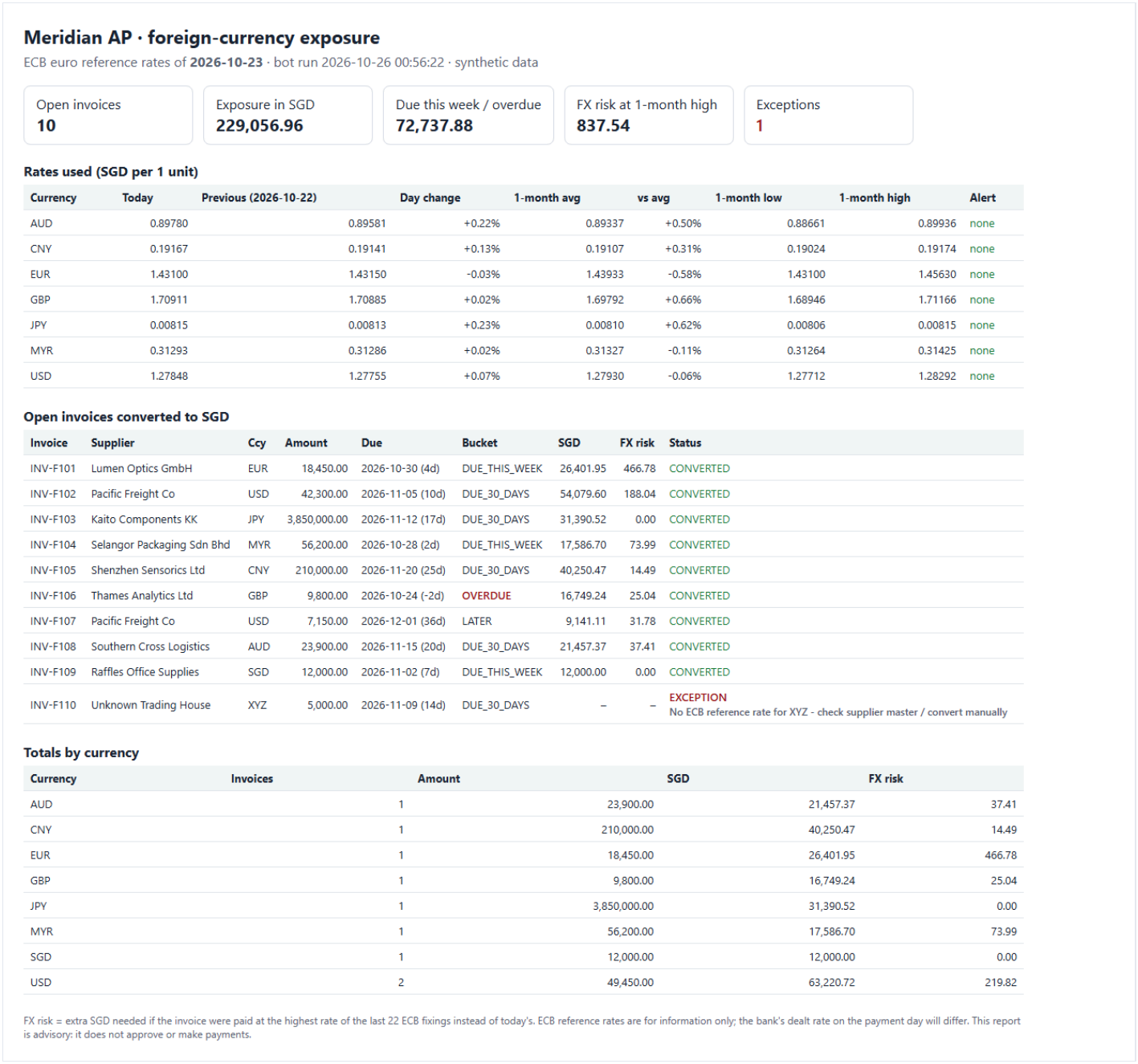
1. **Load inputs.** `input/open_invoices.csv` (10 synthetic invoices in 7 currencies plus one bad code, XYZ) and `input/bot_settings.csv` (alert thresholds the AP team can change without editing the script).
2. **Read today's rates.** Opens the ECB page, declines the cookie banner, reads the reference date heading and, for each currency, the spot rate using XPath anchored on the table row id (`//td[@id="USD"]/../td[contains(@class,"spot")]//span[@class="rate"]`). A currency missing from the table (XYZ) is recorded rather than crashing the run.
3. **Evidence.** Screenshots of the rate table and the whole page.
4. **History per currency.** For each currency the bot clicks through to its chart page and reads the last ~40 daily rates from the page's own chart data (`generateChartData()`, via a `dom` step), then screenshots the chart. The chart script sometimes loads late, so the read is retried up to 5 times; this was added after a real failure during testing.
5. **Cross rates and statistics.** SGD per unit = (EUR→SGD) ÷ (EUR→currency), for today and every historical date. Day-on-day change, 1-month (22 fixings) average, low and high.
6. **History file.** Appends today's rates to `output/fx_rates_history.csv` only if that ECB date is not already there.
7. **Invoices.** Amount in SGD, days to due, bucket (overdue / this week / 30 days / later), and FX risk = extra SGD if paid at the 1-month high.
8. **Alerts.** FX move ≥ 0.5% day-on-day or ≥ 2% from the 1-month average; overdue invoices; unsupported currencies.
9. **Report.** Builds an HTML report, shows it in the bot's browser and screenshots it; writes one line to `output/run_log.csv`.



Step 4: USD chart page captured after the bot clicked through from the rate table.

4. Results

Report produced by the final run



Meridian AP · foreign-currency exposure

ECB euro reference rates of **2026-10-23** · bot run 2026-10-26 00:56:22 · synthetic data

Open invoices	Exposure in SGD	Due this week / overdue	FX risk at 1-month high	Exceptions
10	**229,056.96**	**72,737.88**	**837.54**	**1**

Rates used (SGD per 1 unit)

Currency	Today	Previous (2026-10-22)	Day change	1-month avg	vs avg	1-month low	1-month high	Alert
AUD	0.89780	0.89581	+0.22%	0.89337	+0.50%	0.88661	0.89936	none
CNY	0.19167	0.19141	+0.13%	0.19107	+0.31%	0.19024	0.19174	none
EUR	1.43100	1.43150	-0.03%	1.43933	-0.58%	1.43100	1.45630	none
GBP	1.70911	1.70885	+0.02%	1.69792	+0.66%	1.68946	1.71166	none
JPY	0.00815	0.00813	+0.23%	0.00810	+0.62%	0.00806	0.00815	none
MYR	0.31293	0.31286	+0.02%	0.31327	-0.11%	0.31264	0.31425	none
USD	1.27848	1.27755	+0.07%	1.27930	-0.06%	1.27712	1.28292	none

Open invoices converted to SGD

Invoice	Supplier	Ccy	Amount	Due	Bucket	SGD	FX risk	Status
INV-F101	Lumen Optics GmbH	EUR	18,450.00	2026-10-30 (4d)	DUE_THIS_WEEK	26,401.95	466.78	CONVERTED
INV-F102	Pacific Freight Co	USD	42,300.00	2026-11-05 (10d)	DUE_30_DAYS	54,079.60	188.04	CONVERTED
INV-F103	Kaito Components KK	JPY	3,850,000.00	2026-11-12 (17d)	DUE_30_DAYS	31,390.52	0.00	CONVERTED
INV-F104	Selangor Packaging Sdn Bhd	MYR	56,200.00	2026-10-28 (2d)	DUE_THIS_WEEK	17,586.70	73.99	CONVERTED
INV-F105	Shenzhen Sensorics Ltd	CNY	210,000.00	2026-11-20 (25d)	DUE_30_DAYS	40,250.47	14.49	CONVERTED
INV-F106	Thames Analytics Ltd	GBP	9,800.00	2026-10-24 (-2d)	OVERDUE	16,749.24	25.04	CONVERTED
INV-F107	Pacific Freight Co	USD	7,150.00	2026-12-01 (36d)	LATER	9,141.11	31.78	CONVERTED
INV-F108	Southern Cross Logistics	AUD	23,900.00	2026-11-15 (20d)	DUE_30_DAYS	21,457.37	37.41	CONVERTED
INV-F109	Raffles Office Supplies	SGD	12,000.00	2026-11-02 (7d)	DUE_THIS_WEEK	12,000.00	0.00	CONVERTED
INV-F110	Unknown Trading House	XYZ	5,000.00	2026-11-09 (14d)	DUE_30_DAYS	–	–	EXCEPTION No ECB reference rate for XYZ - check supplier master / convert manually

Totals by currency

Currency	Invoices	Amount	SGD	FX risk
AUD	1	23,900.00	21,457.37	37.41
CNY	1	210,000.00	40,250.47	14.49
EUR	1	18,450.00	26,401.95	466.78
GBP	1	9,800.00	16,749.24	25.04
JPY	1	3,850,000.00	31,390.52	0.00
MYR	1	56,200.00	17,586.70	73.99
SGD	1	12,000.00	12,000.00	0.00
USD	2	49,450.00	63,220.72	219.82

FX risk = extra SGD needed if the invoice were paid at the highest rate of the last 22 ECB fixings instead of today's. ECB reference rates are for information only; the bank's dealt rate on the payment day will differ. This report is advisory: it does not approve or make payments.

Step 9: the bot's report, opened in its browser and captured.

Open invoices in SGD (output/ap_fx_exposure_2026-10-23.csv)

Invoice	Supplier	Ccy	Amount	Due	Bucket	SGD	FX risk	Status
INV-F101	Lumen Optics GmbH	EUR	18,450.00	2026-10-30	DUE_THIS_WEEK	26,401.95	466.78	CONVERTED
INV-F102	Pacific Freight Co	USD	42,300.00	2026-11-05	DUE_30_DAYS	54,079.60	188.04	CONVERTED
INV-F103	Kaito Components KK	JPY	3,850,000.00	2026-11-12	DUE_30_DAYS	31,390.52	0.00	CONVERTED
INV-F104	Selangor Packaging Sdn Bhd	MYR	56,200.00	2026-10-28	DUE_THIS_WEEK	17,586.70	73.99	CONVERTED
INV-F105	Shenzhen Sensorics Ltd	CNY	210,000.00	2026-11-20	DUE_30_DAYS	40,250.47	14.49	CONVERTED
INV-F106	Thames Analytics Ltd	GBP	9,800.00	2026-10-24	OVERDUE	16,749.24	25.04	CONVERTED
INV-F107	Pacific Freight Co	USD	7,150.00	2026-12-01	LATER	9,141.11	31.78	CONVERTED
INV-F108	Southern Cross Logistics	AUD	23,900.00	2026-11-15	DUE_30_DAYS	21,457.37	37.41	CONVERTED
INV-F109	Raffles Office Supplies	SGD	12,000.00	2026-11-02	DUE_THIS_WEEK	12,000.00	0.00	CONVERTED
INV-F110	Unknown Trading House	XYZ	5,000.00	2026-11-09	DUE_30_DAYS	-	-	EXCEPTION

Runs (output/run_log.csv)

Run	Started	Rate date	FX alerts	Exceptions	Exposure SGD	History rows added	Seconds	Result
1: first run	2026-10-26 00:51:47	2026-10-23	0	1	229,056.96	7	81.8	SUCCESS
2: same date again	2026-10-26 00:53:15	2026-10-23	0	1	229,056.96	0	80.8	SUCCESS
3: alert test (0.5%)	2026-10-26 00:54:49	2026-10-23	3	1	229,056.96	0	85.7	SUCCESS
4: final, normal settings	2026-10-26 00:56:22	2026-10-23	0	1	229,056.96	0	80.9	SUCCESS

Run 2 repeated run 1 on the same ECB date and added no history rows, so the bot can be re-run safely after a failure. Run 3 lowered the off-average threshold to 0.5% to exercise the alert branch (the market was calm, so 2% did not trigger):

Invoice	Ccy	Status in run 3	Note
INV-F101	EUR	CONVERTED_FX_ALERT	EUR OFF_1M_AVERAGE (-0.03% d/d)
INV-F103	JPY	CONVERTED_FX_ALERT	JPY OFF_1M_AVERAGE (0.23% d/d)
INV-F106	GBP	CONVERTED_FX_ALERT	GBP OFF_1M_AVERAGE (0.02% d/d)

Meridian AP · foreign-currency exposure

ECB euro reference rates of **2026-10-23** · bot run 2026-10-26 00:54:49 · synthetic data

Open invoices	Exposure in SGD	Due this week / overdue	FX risk at 1-month high	Exceptions
10	**229,056.96**	**72,737.88**	**837.54**	**1**

Rates used (SGD per 1 unit)

Currency	Today	Previous (2026-10-22)	Day change	1-month avg	vs avg	1-month low	1-month high	Alert
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EUR	1.43100	1.43150	-0.03%	1.43933	-0.58%	1.43100	1.45630	OFF_1M_AVERAGE
GBP	1.70911	1.70885	+0.02%	1.69792	+0.66%	1.68946	1.71166	OFF_1M_AVERAGE
JPY	0.00815	0.00813	+0.23%	0.00810	+0.62%	0.00806	0.00815	OFF_1M_AVERAGE
MYR	0.31293	0.31286	+0.02%	0.31327	-0.11%	0.31264	0.31425	none
USD	1.27848	1.27755	+0.07%	1.27930	-0.06%	1.27712	1.28292	none

Open invoices converted to SGD

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INV-F102	Pacific Freight Co	USD	42,300.00	2026-11-05 (10d)	DUE_30_DAYS	54,079.60	188.04	CONVERTED
INV-F103	Kaito Components KK	JPY	3,850,000.00	2026-11-12 (17d)	DUE_30_DAYS	31,390.52	0.00	CONVERTED_FX_ALERT JPY OFF_1M_AVERAGE (0.23% d/d)
INV-F104	Selangor Packaging Sdn Bhd	MYR	56,200.00	2026-10-28 (2d)	DUE_THIS_WEEK	17,586.70	73.99	CONVERTED
INV-F105	Shenzhen Sensorics Ltd	CNY	210,000.00	2026-11-20 (25d)	DUE_30_DAYS	40,250.47	14.49	CONVERTED
INV-F106	Thames Analytics Ltd	GBP	9,800.00	2026-10-24 (-2d)	OVERDUE	16,749.24	25.04	CONVERTED_FX_ALERT GBP OFF_1M_AVERAGE (0.02% d/d)
INV-F107	Pacific Freight Co	USD	7,150.00	2026-12-01 (36d)	LATER	9,141.11	31.78	CONVERTED
INV-F108	Southern Cross Logistics	AUD	23,900.00	2026-11-15 (20d)	DUE_30_DAYS	21,457.37	37.41	CONVERTED
INV-F109	Raffles Office Supplies	SGD	12,000.00	2026-11-02 (7d)	DUE_THIS_WEEK	12,000.00	0.00	CONVERTED
INV-F110	Unknown Trading House	XYZ	5,000.00	2026-11-09 (14d)	DUE_30_DAYS	–	–	EXCEPTION No ECB reference rate for XYZ - check supplier master / convert manually

Totals by currency

Currency	Invoices	Amount	SGD	FX risk
AUD	1	23,900.00	21,457.37	37.41
CNY	1	210,000.00	40,250.47	14.49
EUR	1	18,450.00	26,401.95	466.78
GBP	1	9,800.00	16,749.24	25.04
JPY	1	3,850,000.00	31,390.52	0.00
MYR	1	56,200.00	17,586.70	73.99
SGD	1	12,000.00	12,000.00	0.00
USD	2	49,450.00	63,220.72	219.82

FX risk = extra SGD needed if the invoice were paid at the highest rate of the last 22 ECB fixings instead of today's. ECB reference rates are for information only; the bank's dealt rate on the payment day will differ. This report is advisory: it does not approve or make payments.

Run 3: alert rows highlighted in the report.

Rate history (output/fx_rates_history.csv)

Rate date	Currency	Units per EUR	SGD per unit	Day change %
2026-10-23	AUD	1.5939	0.897798	0.222
2026-10-23	CNY	7.466	0.191669	0.134
2026-10-23	EUR	1	1.431000	-0.035
2026-10-23	GBP	0.83728	1.709106	0.015
2026-10-23	JPY	175.51	0.008153	0.227
2026-10-23	MYR	4.5729	0.312931	0.024
2026-10-23	USD	1.1193	1.278478	0.072

Alerts written (output/fx_alerts.csv)

Logged	Ccy	Type	Detail	Message
2026-10-26 00:51:47	GBP	OVERDUE_INVOICE	INV-F106	Thames Analytics Ltd was due 2026-10-24
2026-10-26 00:51:47	XYZ	INVOICE_EXCEPTION	INV-F110	No ECB reference rate for XYZ - check supplier master / convert manually
2026-10-26 00:53:15	GBP	OVERDUE_INVOICE	INV-F106	Thames Analytics Ltd was due 2026-10-24
2026-10-26 00:53:15	XYZ	INVOICE_EXCEPTION	INV-F110	No ECB reference rate for XYZ - check supplier master / convert manually
2026-10-26 00:54:49	EUR	FX_MOVE	OFF_1M_AVERAGE	SGD per EUR moved -0.03% vs 2026-10-22, -0.58% vs 1-month average
2026-10-26 00:54:49	JPY	FX_MOVE	OFF_1M_AVERAGE	SGD per JPY moved 0.23% vs 2026-10-22, 0.62% vs 1-month average
2026-10-26 00:54:49	GBP	FX_MOVE	OFF_1M_AVERAGE	SGD per GBP moved 0.02% vs 2026-10-22, 0.66% vs 1-month average
2026-10-26 00:54:49	GBP	OVERDUE_INVOICE	INV-F106	Thames Analytics Ltd was due 2026-10-24
2026-10-26 00:54:49	XYZ	INVOICE_EXCEPTION	INV-F110	No ECB reference rate for XYZ - check supplier master / convert manually
2026-10-26 00:56:22	GBP	OVERDUE_INVOICE	INV-F106	Thames Analytics Ltd was due 2026-10-24
2026-10-26 00:56:22	XYZ	INVOICE_EXCEPTION	INV-F110	No ECB reference rate for XYZ - check supplier master / convert manually

5. Is it right? Independent check

The bot's figures come from scraping web pages, so a second, independent route was used to check them: the ECB also publishes the same rates as an XML file for machines. `tools/verify_against_ecb_xml.py` downloads it, recomputes every SGD amount and FX-risk figure, and compares them with the bot's CSV.

```
Checking ap_fx_exposure_2026-10-23.csv against https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist-90d.xml
Reference date 2026-10-23; 1-month window 2026-09-24 .. 2026-10-23 (22 fixings)
```

```
INV-F101  EUR      18,450.00  SGD bot   26,401.95 / feed   26,401.95   risk bot   466.78 / feed   466.78   OK
INV-F102  USD      42,300.00  SGD bot   54,079.60 / feed   54,079.60   risk bot   188.04 / feed   188.04   OK
INV-F103  JPY   3,850,000.00  SGD bot   31,390.52 / feed   31,390.52   risk bot     0.00 / feed     0.00   OK
INV-F104  MYR      56,200.00  SGD bot   17,586.70 / feed   17,586.70   risk bot    73.99 / feed    73.99   OK
INV-F105  CNY     210,000.00  SGD bot   40,250.47 / feed   40,250.47   risk bot    14.49 / feed    14.49   OK
INV-F106  GBP       9,800.00  SGD bot   16,749.24 / feed   16,749.24   risk bot    25.04 / feed    25.04   OK
INV-F107  USD       7,150.00  SGD bot    9,141.11 / feed    9,141.11   risk bot    31.78 / feed    31.78   OK
INV-F108  AUD      23,900.00  SGD bot   21,457.37 / feed   21,457.37   risk bot    37.41 / feed    37.41   OK
INV-F109  SGD      12,000.00  SGD bot   12,000.00 / feed   12,000.00   risk bot     0.00 / feed     0.00   OK
INV-F110  XYZ: not an ECB currency -> bot status EXCEPTION  OK

10 invoices checked, 0 mismatches -> PASS
```

6. Controls and design choices

- **Read-only.** The bot reads a public page and writes local files. No logins, no credentials, no payments.
- **Re-runnable.** History is keyed on ECB date; repeating a run does not duplicate rows (run 2).
- **Fails loudly.** Missing elements stop a `read` step with an error instead of producing a zero; an unknown currency becomes an `EXCEPTION` row for a person to handle.
- **Evidence every run.** Screenshots of the table, page, each chart and the report, plus the console log of each run.
- **Settings outside the code.** Thresholds live in a CSV the AP team can edit.
- **Human decision kept.** The report says it is advisory; FX risk is a planning number, not a hedge instruction.

7. Building it: problems met and fixed

TagUI's last release (v6.110, June 2022) does not run as shipped on Windows 11 25H2 with Chrome 156. Five fixes were needed, now packaged as `tools/patch_tagui_2026.ps1` (tested on the original files, safe to re-run):

1. Chrome rejects TagUI's DevTools websocket without `--remote-allow-origins=*`: TagUI hung at "listening for inputs".
2. `wmic.exe` has been removed from Windows 11; TagUI used it to find and close its Chrome. Replaced with a PowerShell helper.
3. TagUI's PATH check looks for `casperjs` but the Windows package ships `casperjs.exe`: "'casperjs' is not recognized".
4. `find /i` resolved to Git's Unix `find` and scanned the C: drive. Pinned to Windows' `find.exe`.
5. Element screenshots were clipped with viewport coordinates, so anything below the fold came out blank. Switched to page coordinates.

In the flow itself: TagUI wraps each step in its own function, so helper functions had to be global; Chrome blocks a web page from opening a local `file://` report, so the report HTML is written into a blank tab instead; and the chart-data read got a retry loop.

8. Limits and next steps

- ECB reference rates are fixed around 16:00 CET (late evening in Singapore), so a 09:00 SGT run uses the previous business day's rate. They are indicative; the bank's dealt rate on payment day will differ.

- The bot depends on the ECB page layout. A redesign breaks the `read` steps (loudly); the XML check would catch a silent change.
- Next: read invoices straight from the AP system export instead of a CSV, schedule it in Task Scheduler at 09:00, and send the alerts file to the team chat (the Week 10 n8n workflow already has a Telegram channel that could receive it).

Files: `fx_ap_bot.tag` (script), `run_bot.cmd`, `input/`, `output/`, `evidence/`, `tools/`. All invoice data is synthetic; exchange rates are real ECB reference rates.